

Smart Attendance System Using BLE Beacons & Flutter App

Milestone 2: Integrate Attendance Flutter App with Firebase

Objective: Enhance the Flutter app to send/retrieve data from Firebase, implement attendance tracking logic, and refine beacon-based detection.

Software Tools:

- Same as in milestone 1.

Requirements:

1. **Integrate your Flutter App with firebase database:**

- (a) Configure Firebase Realtime Database or Cloud Firestore to store:
 - Student profiles (user UID, name, email).
 - Beacon data (UUID, class/course association, date and time duration).
 - Attendance records (timestamp, student UID, beacon UUID, status: "attended" / "absent").
- (b) Interact with the Firebase database by CRUD operations (Create, Read, Update, Delete)
 - Create Data: Push detected beacon UUIDs, student UUIDs, and calculated distances to Firebase (You may calculate connected period time).
 - Read Data: Fetch attendance history for students (e.g., "Attendance Record" screen).
 - Delete or Update Data : Delete or edit attendance record if you took the attendance of wrong class (entered by mistake). You can think of it as a button that will return you to the home page where you can select another course to take attendance and remove the entry of the mistaken course attendance.

2. **Attendance Logic :**

- (a) Trigger attendance recording only when:
 - Beacon distance $\leq$ threshold (e.g., 2 meters).
 - Student is logged in and class is in session (validate against date and time of the class).
- (b) Status Determination:
 - Attended: Beacon detected within range for $\geq$ 2 minutes e.g. (prevent drive-by false positives for students who leave once taking attendance).
 - Absent: No beacon detected within range during the class time.

3. **Enhanced UI & Features :**

- (a) Attendance Record screen :
 - Show real-time status ("Present/Absent") and history of the selected course attendance.

Team leaders should coordinate the testing of their team's Flutter app to verify Firebase writes/reads using debug logs or Firebase Console. You should test edge cases (e.g., multiple beacons).

Submissions Guidelines :

- You should work with your milestone 1 team.
- The deadline for milestone 1 submission is 13-5-2025 at 11:59 pm.

- Any cheating cases will get a zero.
- you'll need to submit:
 - A working app.
 - Screenshots of the database.
- Only one team member (preferably team leader) should submit the required submissions through this [form](#).

Helping Materials :

- [Android Developer Documentation](#): (Official source for Android app development guides, tutorials, and APIs)
- [Stack Overflow](#): (Vast community forum for programming questions and answers)
- [Flutter Tutorials Center](#): (Documentation for the Flutter and dart programming language widely used in Android development)
- [flutter blue plus library](#): (Popular library for interacting with BLE beacons on Flutter apps)
- [Radius Networks Documentation](#): (Guides and tutorials for their beacon hardware and software platform)
- [Estimote Documentation](#): (Guides and tutorials for their beacon hardware and software platform)
- [BeaconZone](#): (Open-source framework for developing beacon-based apps)
- [Firebase Documentation](#):
- [Firebase Realtime Database or Cloud Firestore](#):
- [Android Authentication with Firebase](#):